creative commons

COMMONS DEED

저작자표시-비영리-변경금지 2.0 대한민국

이용자는 아래의 조건을 따르는 경우에 한하여 자유롭게

- 이 저작물을 복제, 배포, 전송, 전시, 공연 및 방송할 수 있습니다.

다음과 같은 조건을 따라야 합니다:

저작자표시. 귀하는 원저작자를 표시하여야 합니다.

비영리. 귀하는 이 저작물을 영리 목적으로 이용할 수 없습니다.

변경금지. 귀하는 이 저작물을 개작, 변형 또는 가공할 수 없습니다.

- 귀하는, 이 저작물의 재이용이나 배포의 경우, 이 저작물에 적용된 이용허락조건을 명확하게 나타내어야 합니다.
- 저작권자로부터 별도의 허가를 받으면 이러한 조건들은 적용되지 않습니다.

저작권법에 따른 이용자의 권리는 위의 내용에 의하여 영향을 받지 않습니다.

이것은 이용허락규약(Legal Code)을 이해하기 쉽게 요약한 것입니다.

Disclaimer

dCollection

Thesis for the Degree of Master

Development of a microfluidic assay for aspirin resistance of platelets

by

Gyehyu Kim

Department of Mechanical Engineering

Graduate School

Korea University

January, 2017

Thesis for the Degree of Master

Development of a microfluidic assay for aspirin resistance of platelets

Advisor : Prof. Sehyun Shin

by

Gyehyu Kim

A thesis submitted to the Graduate School of the Korea University in partial fulfillment of the requirements for the Degree of Master of Engineering in the Department of Mechanical Engineering

January, 2017

Abstract

Development of a microfluidic assay for aspirin resistance of platelets

Gyehyu Kim

Master of Engineering

Department of Mechanical Engineering

Korea University, Seoul, Korea

Advisor : Prof. Sehyun Shin

As the activation of platelets is found to be the main cause of cardiovascular disease, antiplatelet drugs that suppress the function of platelets have been commonly used. Aspirin is widely prescribed as an antiplatelet drug and inhibits the conversion of arachidonic acid to thromboxane A2. Since patients often show resistance to aspirin, the need for accurate and rapid platelet function tests for appropriate prescriptions is increasing.

In this study, a fibrinogen-coated nylon mesh was placed on the blockage part in the microfluidic chip to bind with the activated platelets. The

microfluidic chip consists of a chamber part that activates the platelets by arachidonic acid, a fibrinogen coated mesh blockage part that reacts with the activated platelets, and a main channel part that allows migration through the blockage part due to vacuum pressure. The surface of nylon 6 mesh was carboxylated by treatment with strong acid(HCl) and glutaraldehyde, and the carboxyl group on the mesh surface was chemically bonded to the fibrinogen by amine coupling. Since fibrinogen is coated on the mesh, the mesh is possible to specifically bind to platelets by reacting with arachidonic acid-activated glycoprotein IIb/IIIa. The activated platelets bound with fibrinogen coated mesh interfere with blood flow by narrowing the channel. Thus, the migration distance, which represents the distance the blood advances within the microfluidic chip, is reduced. As increasing the concentration of arachidonic acid added to blood, the migration distance tended to be shorter by activated platelets. Because in vitro experiments with aspirin were difficult, we used an antagonist called ramatroban, which inhibits pathways such as aspirin. On the other hand, despite the addition of the same concentration of arachidonic acid, a longer migration distance was observed as the concentration of ramatroban added to the blood increased.

We successfully quantified the function of platelets by analyzing the migration distance of blood in microfluidic chips with fibrinogen coated

고려대학교 KOREA UNIVERSITY 1905

mesh. In conclusion, this chip makes it possible to evaluate whether aspirin-induced platelet function is properly inhibited or not by aspirin resistance.

List of Figures

Figure 1 Correlation between aspirin resistance and death, myocardial infarction, or stroke[5] ..... - 6 -

Figure 2 (a) Molecular structure of aspirin, (b)Laboratory assessment of aspirin response[9] ..... - 9 -

Figure 3 Schematic illustration of experimental apparatus. ..... - 14 -

Figure 4 The process of coupling of carboxyl group and amine group by amine coupling using EDC / NHS ..... - 17 -

Figure 5 Results of hemolysis of whole blood according to AA concentration. ..... - 21 -

Figure 6 Microscopic observation of platelets in PRP against AA and ramatroban. (a) Normal PRP. (b) PRP with 0.5mM AA. (c) 1μM of ramatroban treated PRP with 0.5mM AA ..... - 23 -

Figure 7 The result of observation of the mesh reacted with anti-fibrinogen-antibody-FITC by fluorescence microscope. (a) Optic image of bare mesh. (b) Fluorescent image of bare mesh. (c)Fluorescence image of fibrinogen coated mesh. ..... - 25 -

Figure 8 Image histogram obtained from fluorescence image of (a) bare mesh and (b) fib-mesh using image J software. ..... - 28 -

Figure 9 Difference of migration distance according to blockage part conditions .... - 30 -

Figure 10 Graph showing the migration time according to blockage part conditions and (a) absence or (b) presence of AA ..... - 33 -

Figure 11 Actual Flow images in a Microfluidic Chip with the Advance of Blood. (a) after 30sec, (b) after 60sec, (c) after 90sec, and (d) after 120sec after the blood passes through the blockage part ..... - 34 -

Figure 12. Microscopic image of mesh washed with PBS after flow experiment. ..... - 36 -

KOREA UNIVERSITY 1905

Figure 13 Change in the migration distance on a microfluidic chip in accordance with the concentration of AA..... - 38 -

Figure 14. Migration distance at fixed AA concentration for blood treated with ramatroban..... - 41 -

Table of contents

ABSTRACT

LIST OF FIGURES

ABBREVIATIONS AND ACRONYMS

1. INTRODUCTION ..... - 1 -

1.1. IMPORTANCE OF MEASURING ASPIRIN RESISTANCE IN PLATELETS ..... - 1 -

1.2. OBJECTIVES IN THIS STUDY ..... - 3 -

2. BACKGROUND RESEARCH ..... - 4 -

2.1. PLATELETS AND ACTIVATION ..... - 4 -

2.2. RISK OF CARDIOVASCULAR DISEASE ..... - 5 -

2.3. ANTIPLATELET THERAPY AND DRUG RESISTANCE ..... - 7 -

3. MATERIALS & METHODS ..... - 10 -

3.1. BLOOD SAMPLING ..... - 10 -

3.2. MICROFLUIDIC CHIP AND EXPERIMENTAL SETUP ..... - 12 -

3.3. NYLON MESH SURFACE CARBOXYLATION ..... - 15 -

3.4. FIBRINOGEN COATING USING AMINE COUPLING METHOD ..... - 15 -

3.5. FIBRINOGEN COATED MESH FLUORESCENCE STAINING ..... - 18 -

3.6. FIB-MESH COATING EVALUATION BY USING FLUORESCENCE IMAGE PROCESSING ..... - 18 -

4. RESULTS AND DISCUSSIONS ..... - 19 -

4.1. HEMOLYSIS TEST BY AA ..... - 19 -

4.2. RESPONSE OF AGONIST(AA) AND ANTAGONIST(RAMATROBAN) ON PLATELETS ..... - 22 -

4.3. FIBRINOGEN COATED MESH COMPARISON USING FLUORESCENCE MICROSCOPE IMAGES ..... - 24 -

4.3. QUANTITATIVE COMPARISON OF FIBRINOGEN COATING BY IMAGE PROCESSING ..... - 26 -

4.4. DIFFERENCE IN MIGRATION DISTANCE ACCORDING TO BLOCKAGE PART CONDITION ..... - 29 -

4.5. ANALYSIS OF MIGRATION DISTANCE ACCORDING TO BLOCKAGE CONDITION AND PRESENCE OR ABSENCE OF AA ..... - 31 -

KOREA UNIVERSITY 1905

4.6. Validation of binding between activated platelets and fibrinogen coated mesh .. - 35 -
4.7. Effect of concentration of arachidonic acid on migration distance ..... - 37 -
4.8. Migration distance at fixed arachidonic acid concentration for blood treated with ramatroban ..... - 39 -

5. CONCLUSION ..... - 42 -

6. REFERENCE..... - 44 -

고려대학교
KOREA UNIVERSITY
1905

Abbreviations and Acronyms

AA : Arachidonic acid

ASA : Aspirin

COX : Cyclooxygenase

CVD : Cardiovascular disease

EDC : N-(3-Dimethylaminopropyl)-N′-ethylcarbodiimide

GP : Glycol protein

LTA : Light transmittance aggregometer

MD : Migration distance

NHS : N-Hydroxysuccinimide

PFA-200 : Platelet function analyzer-200

PRP : Platelet-rich-plasma

SiPA : Shear induced platelet activation

TD : Touchdown

TXA2 : Thromboxane A2

WB : Whole blood

1. Introduction

1.1. Importance of measuring aspirin resistance in platelets

In the past, platelet function tests were performed mainly to confirm low function of platelets. Because low function of platelets can have a very negative effect on the patient's surgical procedure. However, as platelet overexpression has been identified as a major cause of cardiovascular disease, platelet function tests have been performed predominantly for antiplatelet therapy. VerifyNow and PFA-200 are commonly used in clinical practice. VerifyNow measures the function of platelets by contacting the lyophilized fibrinogen coated beads and agonist with whole blood and reading the turbidity changes due to platelet aggregation. PFA-200 exposes whole blood to a high shear stress condition through a 150μm diameter hollow tube, and measures the occlusion time for aggregated platelets to block blood flow by placing the drug-coated membrane at the exit. They both use disposable chips and have the advantage that no extra labor is required other than the injection of blood. However, both devices are highly dependent on hematocrit, and VerifyNow is very expensive and PFA-200 is dependent on the number of SiPA-related factors called von Willebrand factor.[1]

고려대학교 KOREA UNIVERSITY 1905

In our previous work, we developed a platelet aggregation measuring device capable of quantitative evaluation[2, 3]. The blood was injected into the assembled microfluidic chip to react with the drug, and then the blood flow in the channel was analyzed by sucking it under the negative pressure, so that the platelet function could be evaluated by the difference in the viscosity of blood due to platelet activation.

1.2. Objectives in this study

In this study, the function of platelets can be visually evaluated by analyzing the change of blood flow by binding GP IIb/IIIa of activated platelet surface with immobilized fibrinogen of flow blockage part. A simple fabrication protocol is required for commercialization and a microfluidic chip should be designed so that the user's operation is minimized.

2. Background research

2.1. Platelets and activation

Platelets, called thrombocytes, are one of the cells in our blood. This size in the state not activated, usually from 1 to 3μm and has a structure of a biconvex discoid. The main function of this is to perform a hemostatic action to stop the blood near the damaged endothelial cells. First, platelets bind to damaged and exposed vascular endothelial cells : Adhesion. Second, they change their structure, activate the receptor, and secrete some materials for chemical signaling : Activation. Third, they connect each other through a number of activated receptors and connectors to form a large clot to plug the hole : Aggregation. The formation of these platelet clots are related to the activation process of the cascade that forms the thrombus, which is the result of fibrin accumulation and linking.

Platelet activation can be caused by physical and chemical stimuli. It can be activated by physical stimulation by high shear stress, called shear induced platelet activation (SiPA). Activation by chemical stimulation is called drug induced platelet activation (DiPA), which includes drugs such as epinephrine, ADP, and collagen.

고려대학교 KOREA UNIVERSITY 1905

2.2. Risk of cardiovascular disease

Cardiovascular disease is a category related to the heart or blood vessels, including acute myocardial infarction and stroke. The two diseases mentioned above are accompanied by atherosclerosis, accounting for 80% of all CVD-related deaths4. Atherosclerosis is caused by leukocytes infiltrating and accumulating in the walls of the arterial blood vessels, which causes narrowing of blood vessels. If the formed thrombus blocks the narrowed blood vessels and blocks the flow of blood, the supply of oxygen becomes impossible and the cells become necrotic. It is often difficult to predict because of sudden onset without symptoms, and may lead to death if proper and quick treatment is not done. Therefore, in order to prevent cardiovascular diseases, drug therapy that inhibits platelet function and interferes with thrombus formation has been widely practiced.

고려대학교 KOREA UNIVERSITY 1905

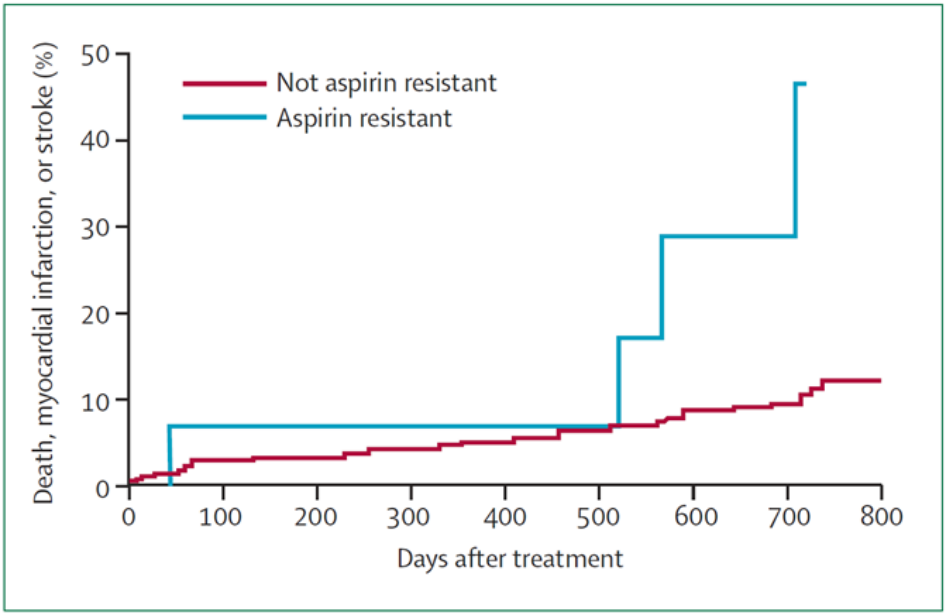


Figure 1 Correlation between aspirin resistance and death, myocardial infarction, or stroke[5]

2.3. Antiplatelet therapy and Drug resistance

Typical antiplatelet therapy drugs include aspirin, clopidogrel, and abciximab, among which aspirin is widely used. Aspirin is a drug called acetylsalicylic acid, was a drug developed for treating pain and fever, but it later turned out to have side effects that reduced the function of platelets irreversibly. By reversing the side effects of aspirin, it has become possible to prevent cardiovascular diseases by lowering the function of platelets. There is a difference between diagnostic devices and groups, but about 20 percent of aspirin users have resistance[5, 6, 7, 8]. Therefore, aspirin prescription as antiplatelet therapy should be done with the diagnosis of aspirin resistance in patients. If the patient has resistance to aspirin, he or she should be prescribed a drug that reduces platelet function by blocking other pathways.

There are several ways to activate platelets with drugs. Typical agonists include adenosine diphosphate (ADP), collagen, and arachidonic acid (AA). Among them, the substance related to aspirin is AA, which is converted into thromboxane A2 (TXA2) by cyclooxygenase (COX-1) enzyme in platelets. The converted TXA2 is released outside of the platelets and binds to the TXA2 receptor of the other platelets, which activates GP IIb/IIIa. GP IIb/IIIa is a type of glycoprotein on the surface of platelets, which forms a flocculant by binding fibrinogen as a linker to activated GP IIb/IIIa on the surface of

고려대학교 KOREA UNIVERSITY 1905

other platelets. Low doses of aspirin prevent the conversion of AA to TAX2 through the irreversible inhibition of COX and consequently inhibit activation of GP llb/llla to prevent platelets from aggregating together.

Ramatroban was originally developed as a treatment for asthma, after which it was found that TXA2 receptor-mediated inhibition prevented TXA2 produced by AA from causing platelet aggregation. In fact, the target that ramatroban acts on is the TXA2 receptor, which is different from aspirin but inhibits the same pathway, resulting in the same effect.

고려대학교
KOREA UNIVERSITY
1905

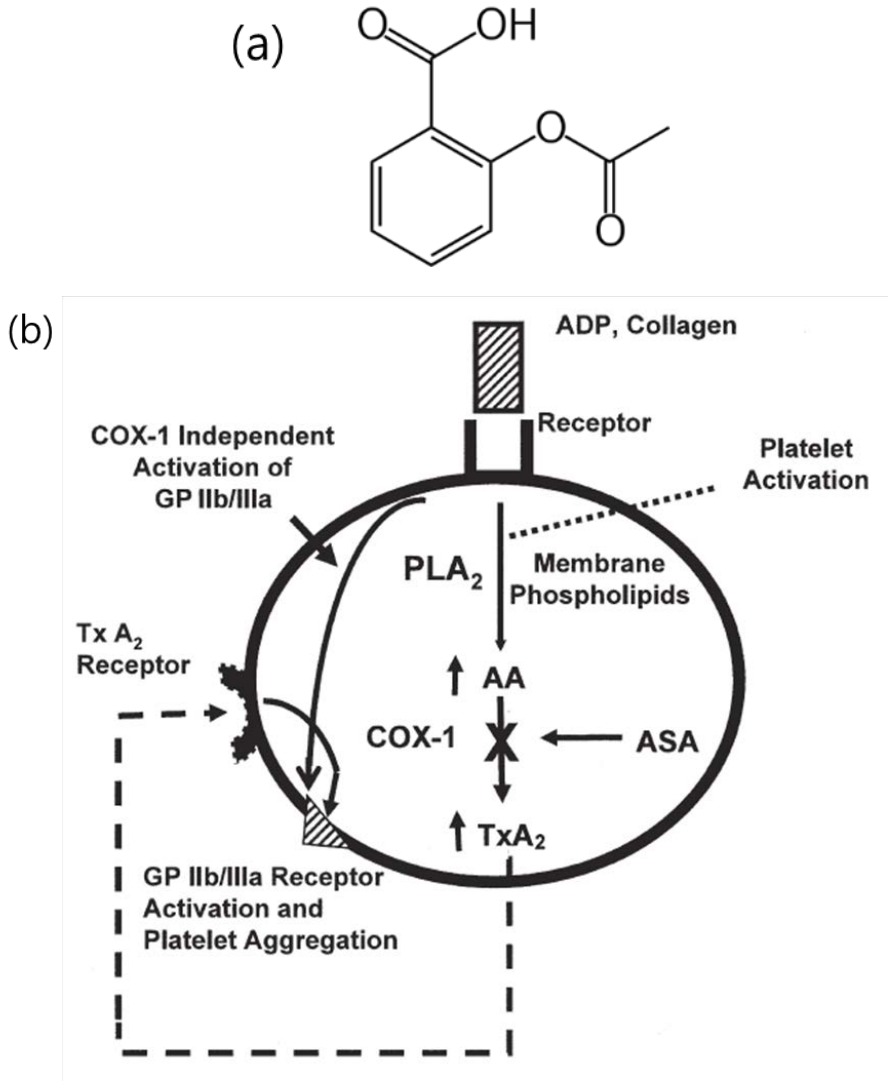


Figure 2 (a) Molecular structure of aspirin, (b)Laboratory assessment of aspirin response[9]

3. Materials & methods

3.1. Blood sampling

For venous whole blood collection, a BD Plastic Vacutainer Plus plastic citrate tube containing 0.109 M sodium citrate coated on a polypropylene wall was used. Considering the change of platelet function with time, sample preparation and all experiments were performed within 4hrs after blood collection[10]. Samples from a healthy donor with no history of aspirin use and no abnormality in platelet function were used. Platelet rich plasma was obtained from the supernatant by centrifugation (180g, 10 min) of whole blood.

To confirm hemolysis of red blood cells according to the concentration of AA, After transferring 500μL of whole blood to a 1.5mL microtube, AA was added at a concentration of 1, 3, 5, 7, 10mM and incubated in a roll mixer for 10 minutes Thereafter, centrifugation was carried out at 180 g for 10mins, and the color of the supernatant of the sample was observed to confirm the occurrence of hemolysis. In flow test with microfluidic chip, AA was used at concentrations of 0.1, 0.5, and 1.0mM, and ramatroban was tested at concentrations of 1, 3, 10, 30, 100, and 300nM with AA fixed at 0.5mM. In the case of ramatroban, the stock dissolved at a concentration of

고려대학교 KOREA UNIVERSITY 1905

5 mg/mL in dimethyl sulfoxide was diluted to 0.1% of the blood volume. For the reaction of ramatroban and blood, they were mixed with whole blood 15-20 min before the experiment and incubated in a roll mixer.[11, 12]

3.2. Microfluidic chip and experimental setup

The microfluidic chip can be divided into a chamber part, a flow blockage part, and a main channel part. The inlet of the acrylic chamber is divided into two chambers in which 400μL of whole blood is injected, one chamber is treated with no treatment, and another chamber is pre-coated with arachidonic acid. The chamber has a cylindrical shape with a diameter of 9 mm and a height of 2 mm and can hold 125μL of blood. Both chambers have magnetic stirrer bars of SUS material with a width of 7 mm, a length of 2mm and a thickness of 0.25mm. The stirrers were rotated at a speed of 300RPM before the start of the blood test, and the whole blood and the drug were stirred. Thereafter, the stirrers were continuously rotated during the experiment to eliminate the effect due to RBC aggregation.

The blockage part was fixed to the lattice-shaped mesh with fibrinogen chemically coated perpendicularly to the direction of blood flow, which caused clogging of activated GP Ilb/IIIa of activated platelets.

Then, in the main channel of long zigzag shape, scale was attached to visually confirm the flow of blood passing through the blockage part. The main channel is made of polytetrafluoroethylene circular hollow tube. The length of the main channel through which the blood passes through the blockage part to the tip of the microfluidic chip is 255mm in total. Therefore,

KOREA UNIVERSITY 1905

the MD can be measured up to 255mm, and no more MD has been measured for blood that exceeds this value.

Connect the assembled microfluidic chip to the vacuum pressure system to measure the pressure generated at the same time as pulling the blood. Since the syringe pulls 0.8mL at a rate of 3.2mL/min, the pressure set in 15 sec is formed, and at the same time, the pressure is measured through the pressure sensor in real time to check for leakage. With the chip lying on the floor, blood was taken from the chamber and advanced to the blockage part and main channel. MD, TD, and TDT were recorded by analyzing images for each experimental condition, and some experiments also analyzed the change of MD according to time, that is, blood velocity.

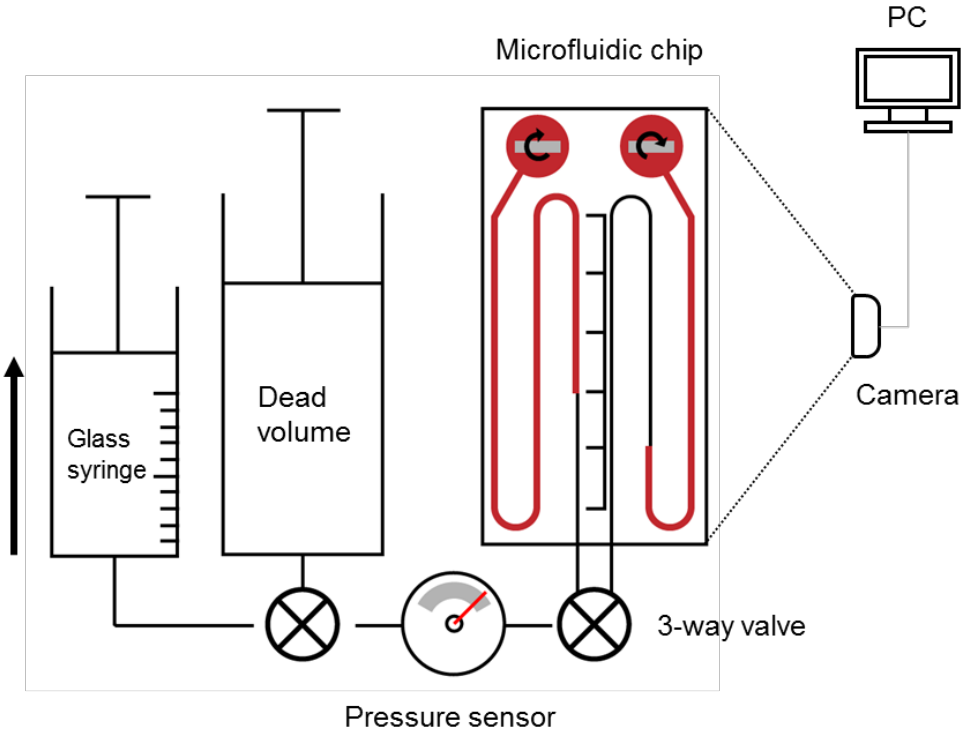


Figure 3 Schematic illustration of experimental apparatus.

3.3. Nylon mesh surface carboxylation

The mesh was partially hydrolyzed by incubation in 6 N hydrochloric acid for 30min and then washed with distilled water. The glutaraldehyde solution diluted to 2.5% in potassium phosphate buffer (pH 6.8) was incubated for 1 hour to carboxylate the surface of the mesh, and then rinsed with potassium phosphate buffer.[13]

3.4. Fibrinogen coating using amine coupling method

Fibrinogen was gently dissolved at a temperature of 37°C with 0.9% saline at a concentration of 2.5mg/mL according to the manual of the data sheet and then filtrated using a sterile-syringe filter of 0.22μm pore size.

Amine coupling is a method of molecular chemistry that allows binding of amine and carboxyl groups of a protein to occur quickly and easily through catalysis of EDC / NHS. As shown in Figure 3, the carboxyl group is converted to O-acylisourea active intermediate by EDC, and the NHS binds to form NHS ester intermediate. When protein is added in this state, amide bonding is formed by bonding with amine group.

For amine coupling, an amine coupling commercial kit from Dojindo was used. The fibrinogen coupling was followed by the product manual but the

고려대학교 KOREA UNIVERSITY 1905

concentration of EDC / NHS solution was increased. Since the mixture of EDC and NHS was not stable, they were dissolved in activation buffer at a concentration of 1M each and mixed immediately before use. Immobilization of proteins by placing a mesh with activated carboxyl groups in the fibrinogen in saline solution for 30 min. After washing the fibrinogen coated mesh with PBS, incubate in blocking buffer containing methanolamine for 30 min to prevent non-specific binding and wash with PBS.[14]

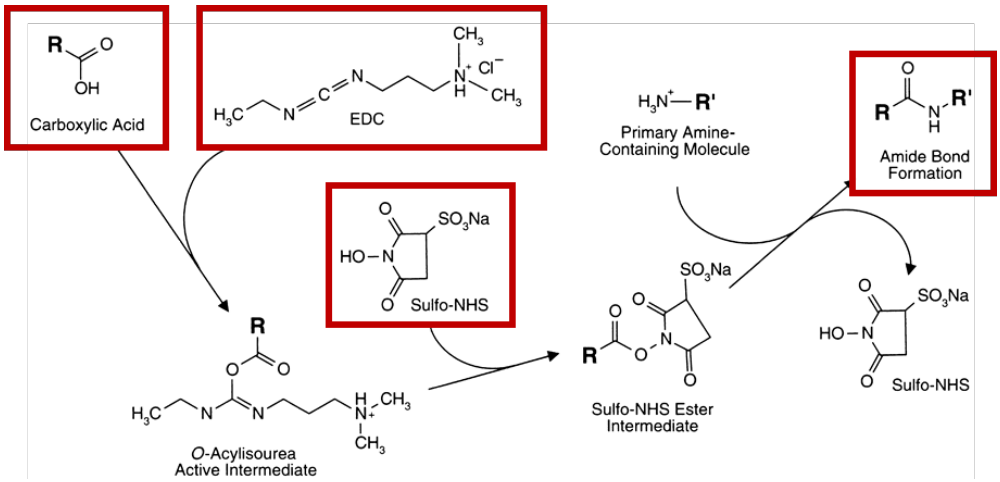


Figure 4 The process of coupling of carboxyl group and amine group by amine coupling using EDC / NHS

3.5. Fibrinogen coated mesh fluorescence staining

Fibrinogen coated mesh(fib-mesh) was incubated in anti-fibrinogen-antibody-FITC solution for 1hr, washed with PBS, cut into appropriate size, placed on slide glass, pressed with cover glass and observed with fluorescence microscope. Incubation, washing, and fluorescence microscopy were performed in the dark room to minimize fluorescence damage caused by natural light.

3.6. Fib-mesh coating evaluation by using fluorescence image processing

Each microscope image was photographed under constant fluorescence exited light source intensity conditions. The image obtained through fluorescence microscopy was processed in image J to obtain a histogram, which is a graph indicating the number of pixels according to the intensity. The fib-mesh coating could be quantified as a median of the intensity of the image from the histogram.

고려대학교
KOREA UNIVERSITY
1905

4. Results and Discussions

4.1. Hemolysis test by AA

Since erythrocyte hemolysis adversely affects the blood test, experiments were conducted to determine the concentration within the range where hemolysis did not occur for the use of AA. When red blood cells are hemolyzed, hemoglobin is eluted outside, and the protein is not settled by centrifugation, so the supernatant floats and becomes reddish. Therefore, it is possible to detect the hemolysis of erythrocytes at each AA concentration by comparing the color of the supernatant.

In the left-most case, only the saline was added without adding AA. When 1mM and 3mM of AA were added, the color of the supernatant was not different from that of the control group as shown in the photograph. However, when 5mM of AA was added, it was confirmed that the supernatant was reddish compared with the control, and it was found that hemolysis occurred. Similarly, in the case of 7mM, the supernatant was reddish, which was relatively redder than the 5mM sample. As the concentration of AA increased, the tendency of more erythrocyte hemolysis occurred at 10mM as well.

고려대학교 KOREA UNIVERSITY 1905

As a result, hemolysis did not occur in 0-3mM concentration of AA, and in case of 5mM or more, hemolysis became worse as concentration increased. Therefore, it was deemed reasonable to use AA within the range of 3mM or less in which erythrocyte hemolysis did not occur. Thus, it was possible to set the concentration of AA suitable for whole blood experiments.

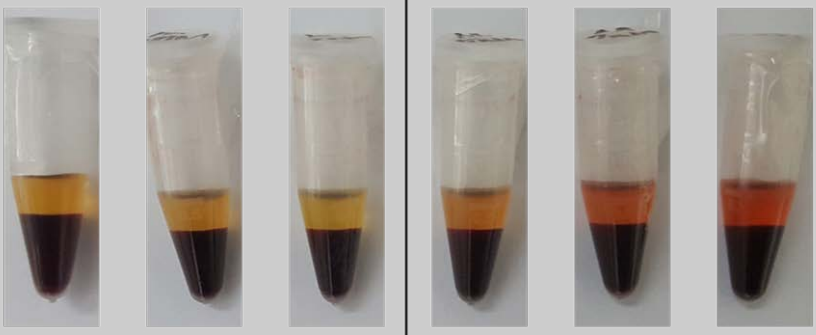
AA[mM]	0	1	3	5	7	10
Images						
Hemolysis	X	X	X	Δ	O	O

Figure 5 Results of hemolysis of whole blood according to AA concentration.

고려대학교 KOREA UNIVERSITY 1905

4.2. Response of agonist(AA) and antagonist(ramatroban) on platelets

To confirm whether AA is activating platelet activation and its action can be inhibited by ramatroban treatment, we performed microscopic observation using PRP.

As shown in Figure 5 (a), platelets are scattered and distributed in the case of normal PRP. Figure 5 (b) shows the addition of 0.5mM of AA, which is an agglutinated platelet aggregate with activated black arrows on the right. The object pointed by the white arrow on the left is red blood cells, which makes it possible to guess the size of aggregated platelets. It can be seen from the scale bar, however, it can be confirmed that it has a size of about 20μm. Considering that the size of the platelets is about 1-3μm, it can be seen that a considerable number of platelets aggregate. Figure 5 (c) shows PRP with activation pathway blocked by AA by adding 1000nM ramatroban. As in Figure 5 (b), 0.5mM AA was added, but no aggregated platelets were found. Thus, 0.5mM AA was sufficient to activate platelets, and 1000nM ramatroban was sufficient to interfere with the action of AA.

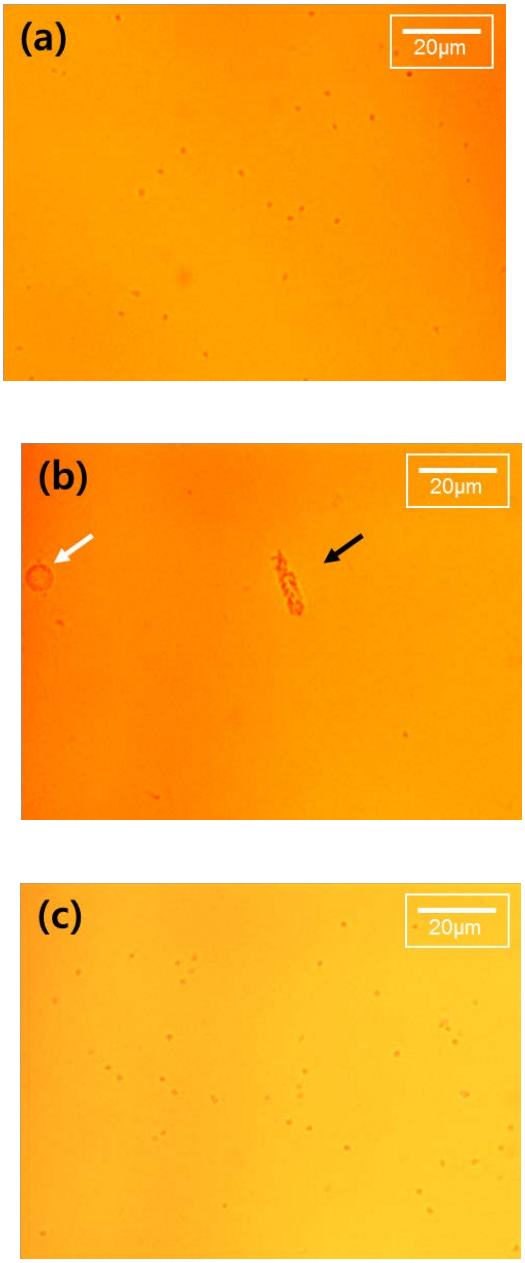


Figure 6 Microscopic observation of platelets in PRP against AA and ramatroban. (a) Normal PRP. (b) PRP with 0.5mM AA. (c) 1μM of ramatroban treated PRP with 0.5mM AA

KOREA UNIVERSITY 1905

4.3. Fibrinogen coated mesh comparison using fluorescence microscope images

In our microfluidic chip, the fibrinogen coated mesh of the blocking part is a very important part designed specifically for the activation of platelets to AA. Therefore, it is essential to confirm that fibrinogen is properly attached to the mesh. Figure 7 shows a microscopic image of the fibrinogen coating of the mesh. Figure 7 (a) is an image of a bare mesh taken by optic, and Figure 7 (b) is an image of the same bare mesh taken in fluorescence in the same composition. This means that the anti-fibrinogen-antibody-FITC does not react because of no fibrinogen immobilized on the surface of the bare mesh and thus is not fluorescent. On the other hand, in the case of Figure 7 (c), since the fibrinogen is immobilized on the surface via carboxylation after the carboxylation of the mesh, it can be reacted with anti-fibrinogen-antibody-FITC to show fluorescence.

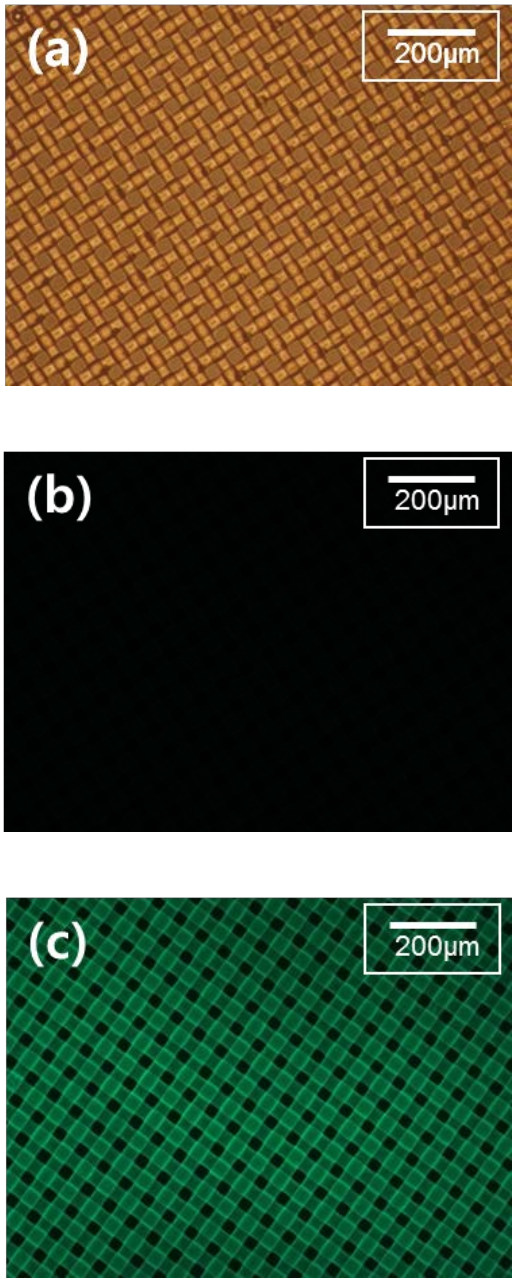


Figure 7 The result of observation of the mesh reacted with anti-fibrinogen-antibody-FITC by fluorescence microscope. (a) Optic image of bare mesh. (b) Fluorescent image of bare mesh. (c)Fluorescence image of fibrinogen coated mesh.

KOREA UNIVERSITY 1905

4.3. Quantitative comparison of fibrinogen coating by image processing

To quantitatively analyze the fluorescence of the previously captured mesh, histograms were obtained by image processing through Image J software. Images are divided into small units called pixels. Since the fluorescent images used are 16-bit images, each pixel is converted to grayscale and has an intensity value between 0 and 255. 0 represents the intensity corresponding to the test, and 255 represents the intensity corresponding to white. The histogram of the image is a graph showing the number of pixels corresponding to each intensity value. In the histogram obtained, the mean value is used as the representative value of the image, which is the sum of the number of pixels corresponding to each intensity and divided by the total number of pixels.

Figure 8 (a) and (b) are histograms obtained from the fluorescence images of Figure 7 (b) and (c), respectively. Figure 8 (a) shows that the number of pixels is concentrated at low intensity. As shown in Figure 7 (b), the image is entirely black. On the other hand, Figure 8 (b) shows that there are many pixels with relatively bright intensity. As can be seen in Figure 7 (c), the pores of the mesh are expressed as black because there is nothing, and the surface of the mesh stained with anti-fibrinogen-antibody-FITC

KOREA UNIVERSITY 1905

emits fluorescence.

The fluorescence of the bare mesh showed a mean value of 2.5 and the fluorescence of the fibrinogen coated mesh had a mean value of 46.8. In the quantitative comparison, it was confirmed that the fibrinogen was coupled with the nylon mesh.

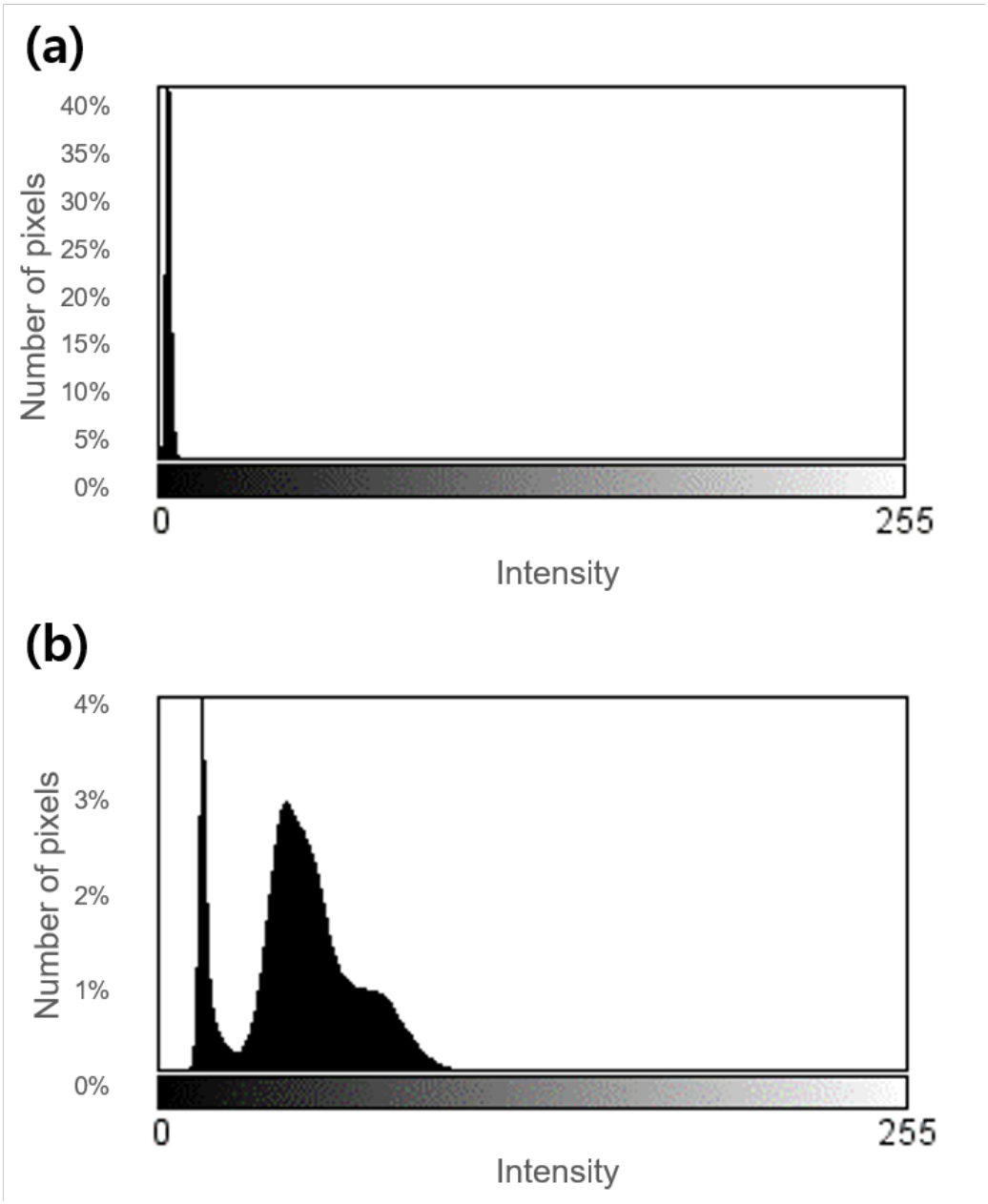


Figure 8 Image histogram obtained from fluorescence image of (a) bare mesh and (b) fib-mesh using image J software.

4.4. Difference in migration distance according to blockage part condition

MD graph based on blockage part status. If no mesh was used (empty), both the untreated control WB and the 0.5mM AA added WB were both touch downed and the time (TDT) required was almost the same. When two bare meshes are placed on the blockage part, the occurrence of TD occurs randomly [223±56]. When two sheets of fibrinogen coated mesh were inserted, the MD was relatively short [68±23].

This means that activated platelets specifically bind to the fibrinogen coated mesh to block the blockage part. As a result, it was confirmed that GP IIb/IIIa of platelet was activated by AA and adhered to fibrinogen coated mesh of blockage part of microfluidic chip to prevent flow.

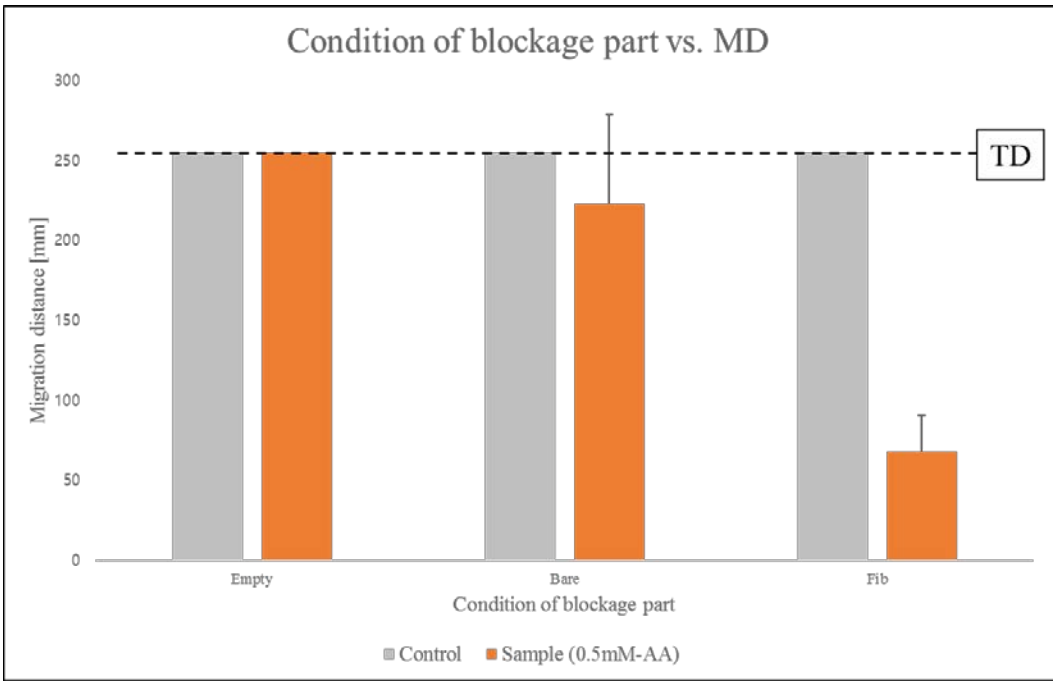


Figure 9 Difference of migration distance according to blockage part conditions

4.5. Analysis of migration distance according to blockage condition and presence or absence of AA

Figure 10 shows the analysis of the MD change over time by the flow analysis in the experiment using the microfluidic chip. The graph above shows the MD for the bare mesh, fibrinogen coated mesh, and the blockage part without the AA added. Regardless of the coating, it can be seen that the flow is slowed down only by using a mesh. This means that the mesh itself acts as a physical resistance. There is no significant difference between bare mesh and fibrinogen coated mesh, indicating that the mesh of the blockage part does not play a large role in the non-activated platelets

The graph below shows the difference of MD according to the composition of the blockage part after adding 0.5mM of AA. In the absence of mesh, similar flow was observed regardless of whether platelet activation by AA was present or not. However, when the mesh was used, TD occurred and showed a large difference. In addition, the fibrinogen coated mesh showed shorter and faster clogging than the bare mesh. Likewise, it can be seen that the fibrinogen-specific GP IIb/IIIa reacted specifically to block the blockage part.

The blood flowing in the upper channel of the chip corresponds to the

고려대학교 KOREA UNIVERSITY 1905

control without addition of AA, and the blood flowing in the lower channel corresponds to the blood having activated platelets added with 0.5mM of AA. After 30sec of advancement, it progresses similarly regardless of whether there is AA or not, and it can be confirmed that the blood of the lower channel has not moved forward since then.

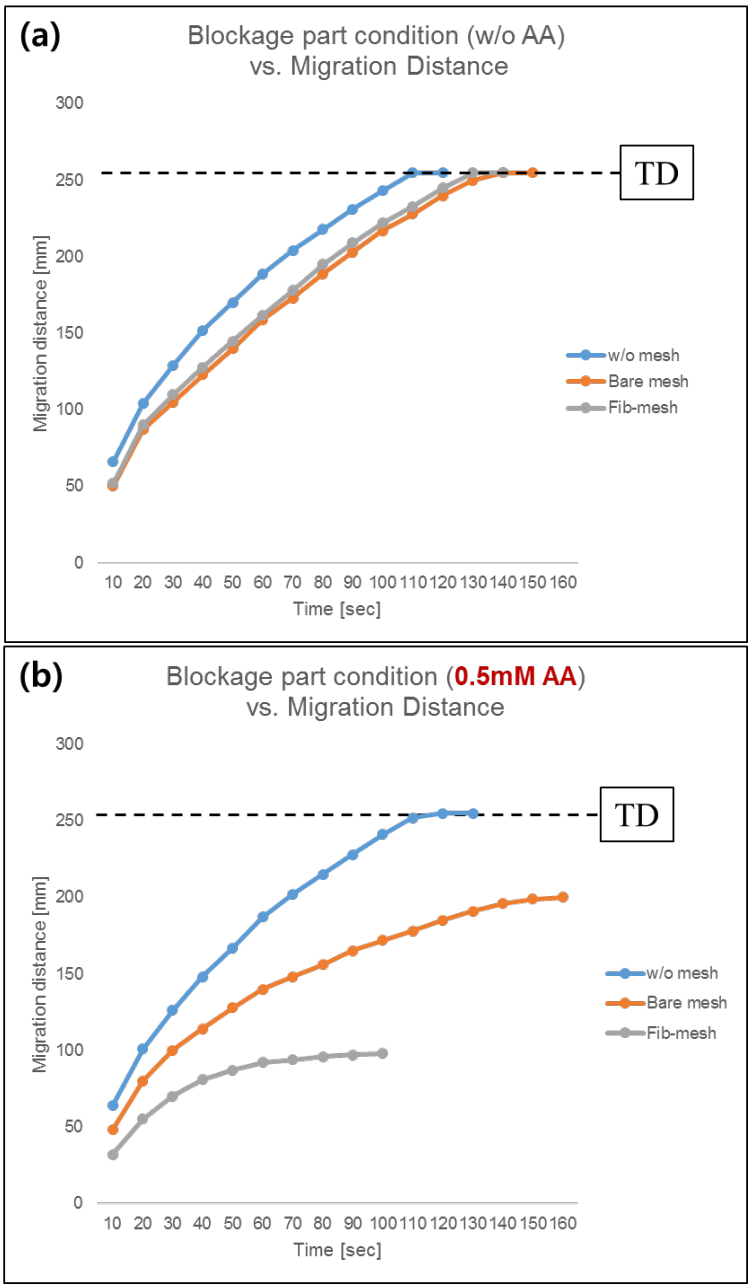


Figure 10 Graph showing the migration time according to blockage part conditions and (a) absence or (b) presence of AA

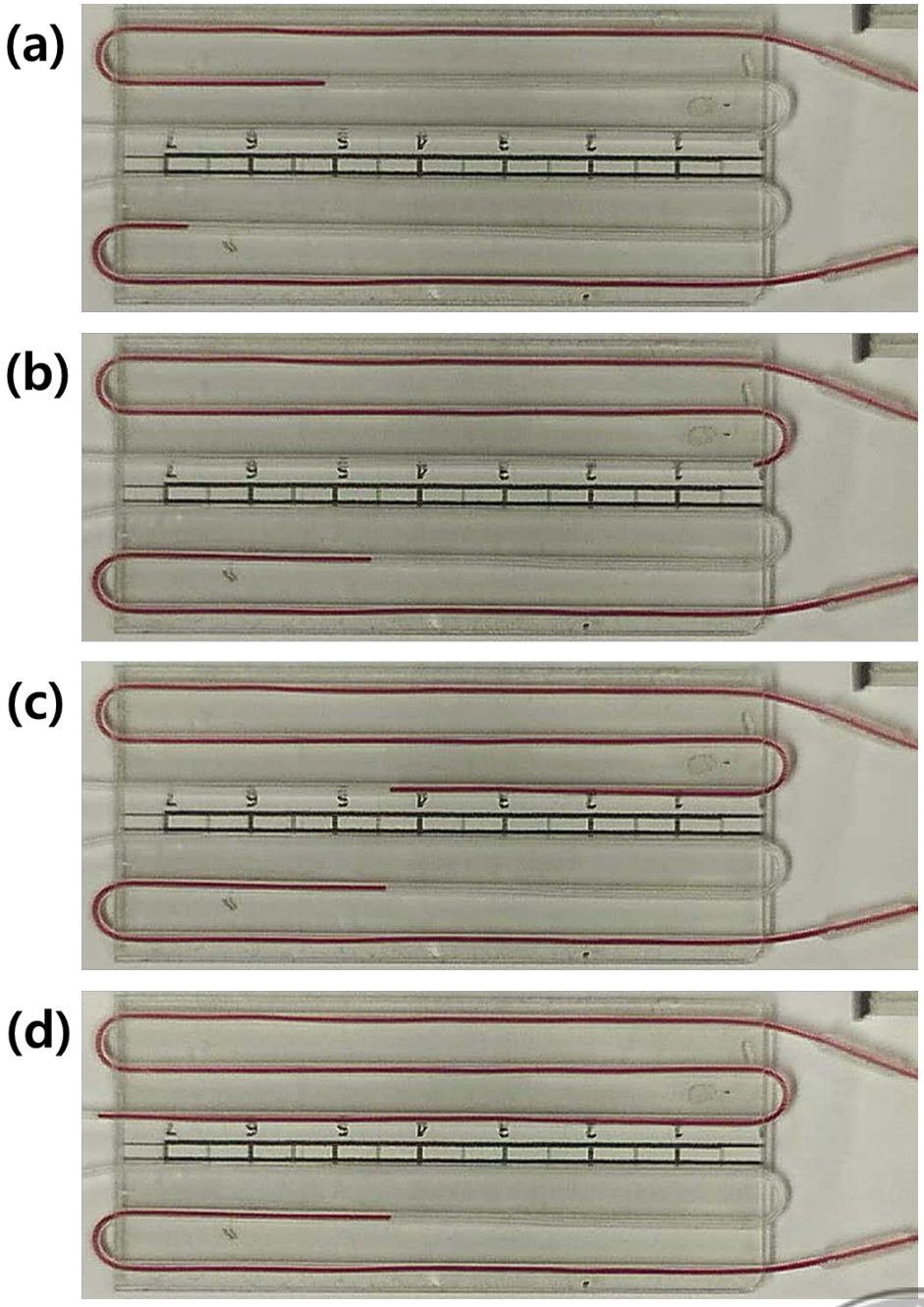


Figure 11 Actual Flow images in a Microfluidic Chip with the Advance of Blood. (a) after 30sec, (b) after 60sec, (c) after 90sec, and (d) after 120sec after the blood passes through the blockage part.

고려대학교 KOREA UNIVERSITY 1905

4.6. Validation of binding between activated platelets and fibrinogen coated mesh

It is an experiment to confirm whether the platelets activated by AA actually reacted to the fibrinogen coated mesh to prevent flow. The first of the two meshes was washed with PBS after blood test and observed with a microscope. When 0.5mM of AA was added and fibrinogen coated mesh was used in the blockage part, it was confirmed that translucent materials stick to the mesh. This is because GP IIb/IIIa of activated platelets aggregates with fibrinogen on the mesh surface. However, there was a case of touchdown even when using bare mesh. Platelets were not attached on the bare mesh obtained after the experiment. Therefore, even in the case of the bare mesh, it is believed that the touchdown occurred because the platelets activated and aggregated by the AA physically blocked the pores of the mesh. Since it is not chemically bound, it is assumed to be washed away during washing with PBS.

To validate this assumption, the fluorescence-stained platelets should be used to observe the mesh with a fluorescence microscope.

고려대학교 KOREA UNIVERSITY 1905

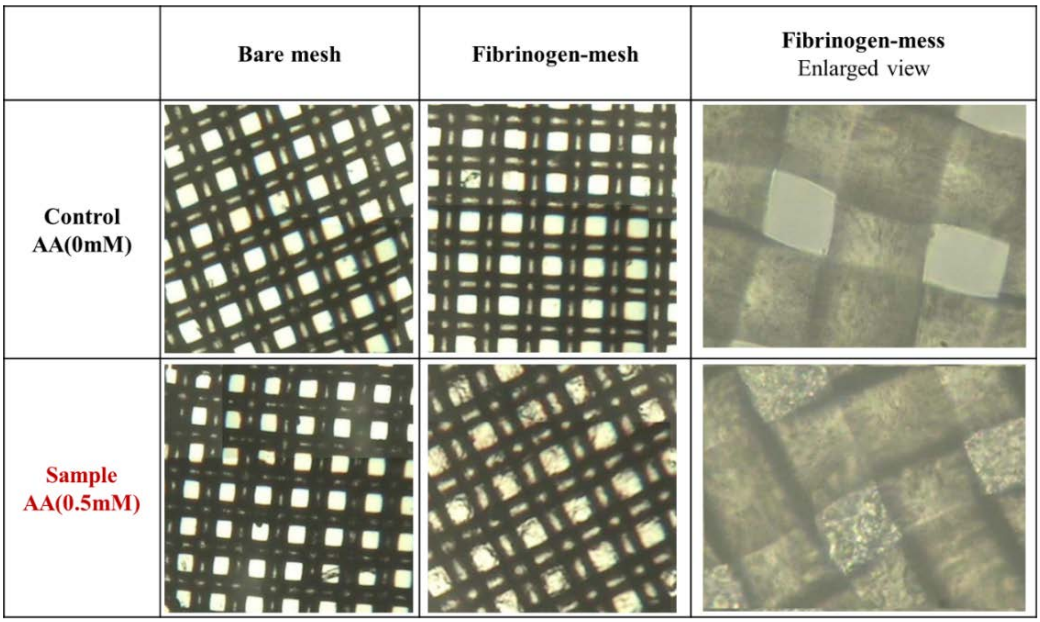


Figure 12. Microscopic image of mesh washed with PBS after flow experiment.

4.7. Effect of concentration of arachidonic acid on migration distance

Figure 13 shows the change of MD according to the concentration of AA. The control group without AA was expressed as 0mM, and the case with AA was divided into 0.1, 0.25, 0.5 and 1.0mM according to the concentration. First, the control group was all touchdown, so the value was equal to 255 and the STD was zero. The MD, STD and CV values according to AA concentration were 255 ± 0 (0%) for 0.1mM, 86 ± 65 (76%) for 0.25mM, 68 ± 23 (33%) for 0.5mM and 19 ± 4 (21%). It was confirmed that the MD was shortened from 0.1mM AA concentration, but because the STD value was relatively large compared to MD (high CV%), we selected 0.5mM AA with appropriate MD and reproducibility. In the case of 1mM, although the CV value was low, it was judged that it was not suitable for the experiment because it had an MD that was too short compared to the entire channel length.

고려대학교 KOREA UNIVERSITY 1905

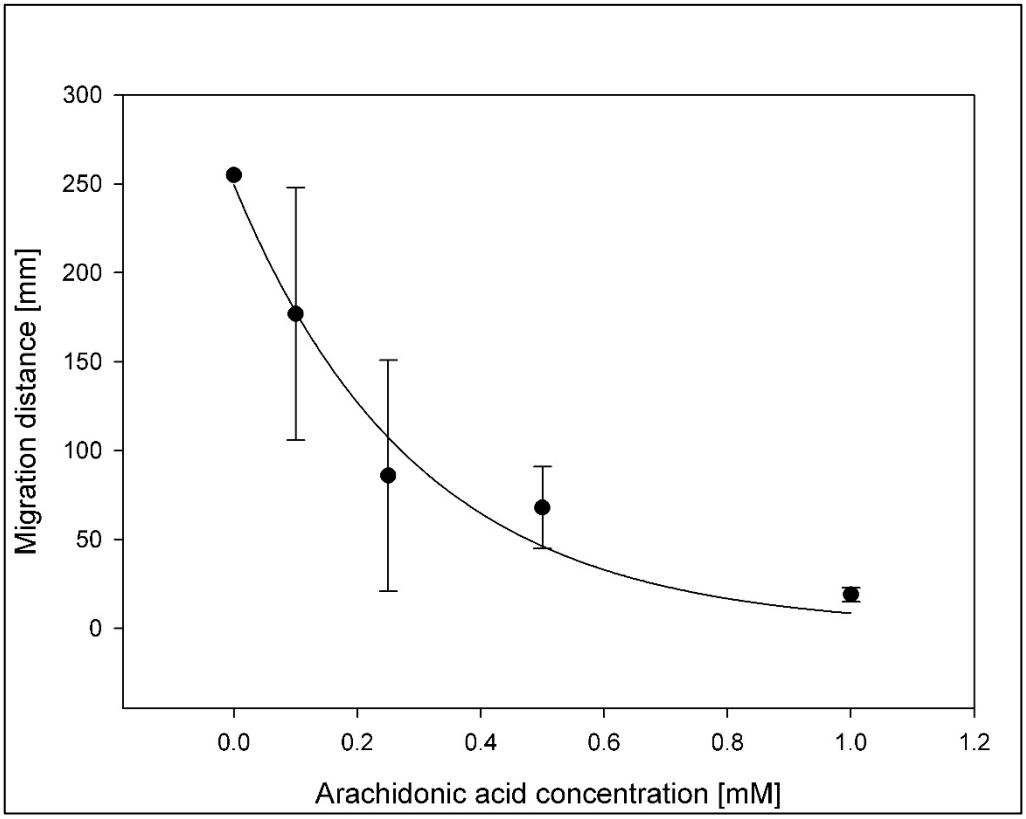


Figure 13 Change in the migration distance on a microfluidic chip in accordance with the concentration of AA

고려대학교 KOREA UNIVERSITY 1905

4.8. Migration distance at fixed arachidonic acid concentration for blood treated with ramatroban

Figure 14 is a graph comparing the MD on the microfluidic chip by adding 0.5mM AA to a blood sample inhibiting the conversion of AA to TXA2 by treatment with ramatroban in blood by each concentration. Since both aspirin and ramatroban inhibit the pathway that activates GP IIb/IIIa, it is necessary to premise that it is possible to diagnose aspirin-resistant individuals through experiments on blood treated with ramatroban. The concentration of nM was used because ramatroban was a potent platelet function inhibitor. Ramatroban was dissolved in DMSO and DMSO actually reduced platelet function. However, in this experiment, only 0.1% of the whole blood volume was used, so it is expected that there will be less than 3% platelet inhibition[12]. As a result, activation by AA was effectively inhibited by ramatroban treatment, which could be confirmed by changes in MD. However, it may be difficult to assess the aspirin resistance step by step because of the extreme touchdown at a low concentration of 100nM, which can be attributed to the fact that the DMSO used for the treatment of ramatroban in the experiment suppressed platelet function. Therefore, if blood of patients taking actual aspirin is used rather than the use of ramatroban and DMSO, consideration of inhibition of platelet function by DMSO can be overcome. The regression line uses the sigmoidal formula,

고려대학교 KOREA UNIVERSITY 1905

$$f = -252.1 + \frac{507.1}{(1+e^{\left(\frac{-(x+0.7)}{12.7}\right)})} \quad (1)$$

, and the R value is 0.9997.

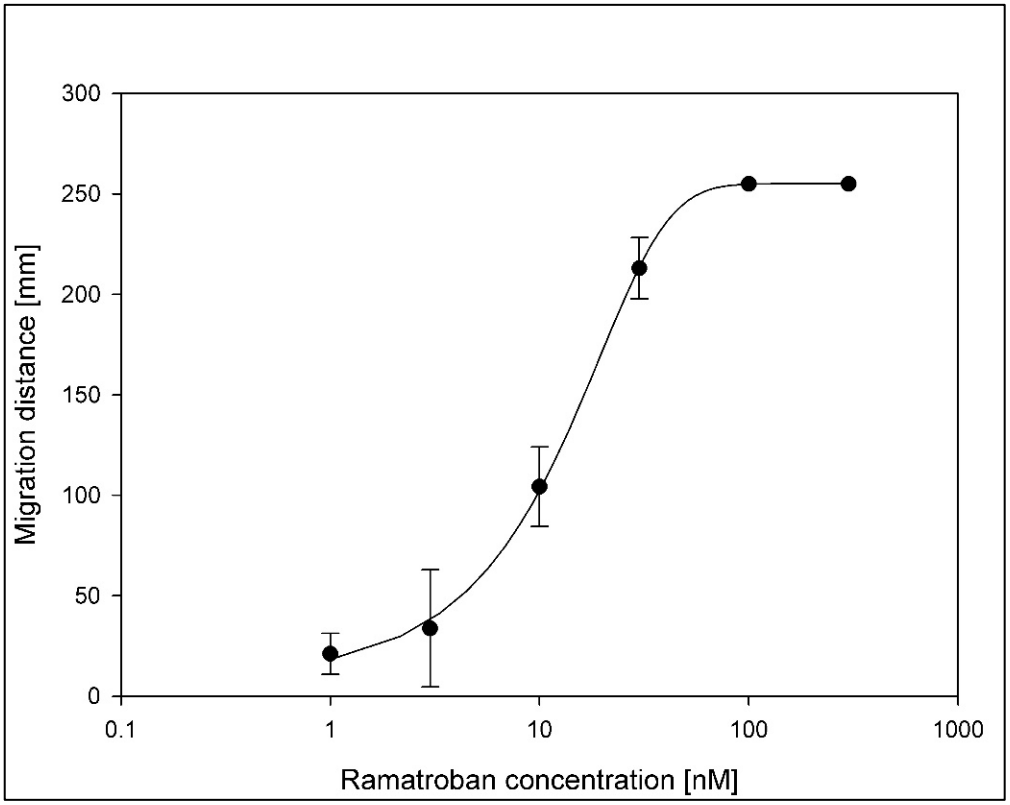


Figure 14. Migration distance at fixed AA concentration for blood treated with ramatroban.

고려대학교
KOREA UNIVERSITY
1905

5. Conclusion

In summary, we have developed a microfluidic assay that can quantitatively measure the function of platelets according to the resistance of aspirin users. Hemolysis test was performed to determine the concentration of AA suitable for the experiment to be less than 3mM and to confirm the platelet activation when 0.5mM AA was added. On the other hand, it was also observed that activation of AA by ramatroban with 1000nM was inhibited in spite of adding the same concentration of AA. Nylon 6 mesh composing the blockage part of the microfluidic chip was treated with hydrolyzation and glutaraldehyde to form carboxyl groups on the surface. The fibrinogen binding to GP IIb/IIIa was coated on the carboxylated mesh through amine coupling and designed to specifically react with GP IIb/IIIa activated by AA. The anti-fibrinogen-antibody-FITC, which specifically binds to fibrinogen, was reacted with fib-mesh and observed by fluorescence microscopy. Fluorescence images were processed with ImageJ which is image processing software to quantitatively compare the samples and confirm that the fibrinogen was successfully coupled to the mesh. In the microfluidic chip test, all inactivated platelets were all touchdown in the absence of mesh, bare mesh, and fib-mesh. However, activated platelets were observed to touch only when there was

KOREA UNIVERSITY 1905

no mesh, and MD with short fib-mesh rather than bare mesh showed that platelets specifically bind to clogging and interfere with flow. After the test, the microscopic observation of the mesh of the chip blockage part revealed that only activated platelets were specifically bound to the fib-mesh and blocked the flow. As the AA concentration increased, the MD decreased exponentially. Finally, in the microfluidic chip test with ramatroban added to the same 0.5mM AA, MD was increased with increasing concentration of antagonist. In the case of 100nM or more, all of them were touchdown. At the concentrations below, the sigmoidal regression line was shown when the x axis antagonist concentration was log scale. Therefore, we have demonstrated the possibility of detecting resistance to antiplatelet agents by quantitatively measuring platelet activation through our microfluidic assay. We believe that testing the resistance to antiplatelet agents that inhibit other pathways such as clopidogrel and ticagrelor as well as aspirin would be possible by setting up differently coated proteins and agonists to activate platelets.

6. Reference

1. Paniccia, R.; Priora, R.; Liotta, A. A.; Abbate, R. Platelet function tests: a comparative review. *Vascular health and risk management* **2014,** *11*, 133-148.

2. Song, S.-H.; Lim, C.-S.; Shin, S. Scalable evaluation of platelet aggregation by the degree of blood migration. *Applied Physics Letters* **2013,** *103* (24), 243702.

3. Song, S.-H.; Lim, C.-S.; Shin, S. Migration distance-based platelet function analysis in a microfluidic system. *Biomicrofluidics* **2013,** *7* (6), 064101.

4. Mendis, S.; Puska, P.; Norrving, B. *Global atlas on cardiovascular disease prevention and control*; World Health Organization2011.

5. Hankey, G. J.; Eikelboom, J. W. Aspirin resistance. *The Lancet* **2006,** *367* (9510), 606-617.

6. Habizal, N. H.; Halim, S. A.; Bhaskar, S.; Bebakar, W. M. W.; Abdullah, J. M. Prevalence of aspirin resistance in diabetic patients and its associated factors. *The Malaysian journal of medical sciences: MJMS* **2015,** *22* (1), 50.

7. Hankey, G. J.; Eikelboom, J. W. Aspirin resistance. *Bmj* **2004,** *328* (7438), 477.

8. Tran, H. A.; Anand, S. S.; Hankey, G. J.; Eikelboom, J. W. Aspirin resistance.

고려대학교 KOREA UNIVERSITY 1905

Thrombosis research **2007,** *120* (3), 337-346.

9. Gurbel, P. A.; Becker, R. C.; Mann, K. G.; Steinhubl, S. R.; Michelson, A. D. Platelet function monitoring in patients with coronary artery disease. *Journal of the American College of Cardiology* **2007,** *50* (19), 1822-1834.

10. Baskurt, O.; Boynard, M.; Cokelet, G.; Connes, P.; Cooke, B. M.; Forconi, S.; Liao, F.; Hardeman, M.; Jung, F.; Meiselman, H. New guidelines for hemorheological laboratory techniques. *Clinical hemorheology and microcirculation* **2009,** *42* (2), 75--97.

11. Chakraborty, R.; Bhullar, R. P.; Dakshinamurti, S.; Hwa, J.; Chelikani, P. Inverse agonism of SQ 29,548 and Ramatroban on thromboxane A2 receptor. *PloS one* **2014,** *9* (1), e85937.

12. Ishizuka, T.; Matsui, T.; Okamoto, Y.; Ohta, A.; Shichijo, M. Ramatroban (BAY u 3405): a novel dual antagonist of TXA2 receptor and CRTh2, a newly identified prostaglandin D2 receptor. *Cardiovasc Drug Rev* **2004,** *22* (2), 71-90.

13. Pahujani, S.; Kanwar, S. S.; Chauhan, G.; Gupta, R. Glutaraldehyde activation of polymer Nylon-6 for lipase immobilization: enzyme characteristics and stability. *Bioresource Technology* **2008,** *99* (7), 2566-2570.

14. Hermanson, G. T. Zero-Length Cross-linkers-3. **1995**.

고려대학교 KOREA UNIVERSITY 1905

ACKNOWLEDGEMENT

석사과정을 마치는 동안 연구적인 고민과 어려움을 해결할 수 있도록 많은 지도와 조언을 주신 신세현 교수님께 깊은 감사를 드립니다. 또한 학위논문 심사에 참여해주신 나성수 교수님과 한창수 교수님께도 감사의 말씀을 올립니다.

학업을 무사히 마칠 수 있도록 물심양면으로 도움을 아끼지 않으시고 격려해주신 부모님께 정말 감사드립니다. 또한 2년동안 제 룸메이트이자 친구, 연구실 동료로서 많은 즐거움과 위로를 주었던 하나뿐인 내 동생 남준이에게도 고마움을 전하고 싶습니다.

나의 정신적 지주 호윤이형, 처음 연구실에 왔을 때부터 졸업을 하는 이 순간까지 저를 이끌어주고 격려해주어 고맙습니다. 당신이 있었기에 저의 석사과정을 무사히 마칠 수 있었다고 생각합니다. 마지막으로 장대호 박사님과 동기 연수, 그리고 연구실의 선후배님들에게도 많은 도움을 준 것에 감사를 드립니다.

2017년 1월 창의관 334호에서 학위 논문을 마치며.

I hereby recommend that the thesis prepared under my supervision by Gyehyu Kim be accepted in partial fulfillment of the requirements for the Degree of Master of Engineering in the Department of Mechanical Engineering

Chair of committee :

Sehyun Shin

Member of committee :

Sungsoo Na

Chang-Soo Han

January 2017